

Project 1

DATA 422: Advanced Data Science Methods and Ethics

Due (full package): Friday, October 2

Weight: 20% of the course (stepping stones count toward this project, not toward labs)

Wildfire insurance AI lab: a **risk model** (Model 1) and later an **evidence-to-text** model (Model 2). Weeks 1–2 are only Model 1.

Target (do not loosen this language)

We estimate

$$\mathbb{P}(\text{serious_damage} = 1 \mid \text{structure inspected in or near a wildfire perimeter})$$

using de-identified CAL FIRE DINS construction features (and later, optional location-derived hazard layers).

serious_damage is 1 for Major or Destroyed, 0 for No Damage / Affected / Minor. Inaccessible rows are out of the supervised table.

This is **not** an annual household loss probability, **not** a premium, and **not** a coverage decision. A separate rule (threshold / abstain / specialist review) will turn scores into an action. Model 2 may describe that action; it may not invent coverage.

Stepping stones

Stone	Due	What you turn in
Week 1	Wed Sep 2, 11:59 p.m.	Contract + PyTorch logistic baseline

Model 1 in PyTorch

Same loop as *The Hundred-Page Language Models Book* and Karpathy:

```
optimizer.zero_grad()
logits = model(x)           # z, not probabilities
loss = nn.BCEWithLogitsLoss()(logits, y)
loss.backward()
optimizer.step()
```

- **Week 1:** `nn.Linear(n_features, 1)` — logistic, $z = \mathbf{w}^\top \mathbf{x} + b$.
- **Week 2:** a small MLP (RiskMLP) with the same loss.

Do not use sklearn for the graded Model 1 fits. Encoding categoricals with pandas is fine.

Data

You will receive a de-identified table (`properties_modeling.csv`) with a `split` column (`train` / `validation` / `test`). Fit on train, make choices on validation, leave test until the final package.

The instructor table is built from public [CAL FIRE DINS](#). Street, APN, and coordinates are stripped before you see it.

Labs vs this project

Tuesday labs are ungraded **Colab** notebooks (practice for the Friday **paper** quiz). They are not stepping stones. The stones on this page are the project checkpoints.